Emilia Ma

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Education

University of Toronto Sept 2024 – Expected May 2029
B.Sc. in Computer Science — Dean's List Scholar (2025, 2026) GPA: 3.76/4.0
- Coursework: Data Structures & Algorithms, Software Design, Human Computer Interaction, Systems Programming, Computer Systems & Digital Logic, Theory of Computation, Probability & Statistics

Experience

Ontario Public Service | Backend Dev/Junior Technical Analyst Sept 2026 – Present
- Restored the order orchestrator's backend test suite from 19 failing scripts to 0 across multiple workflows, by implementing configuration aware test-runner logic to prevent false failures from mocked IDs.
- Implemented Java backend fixes for driver licence applications, adapting the PDF generation to a new template and correcting business date logic to match order records; added JUnit tests for both changes.
- Developed fault-injection profiles and JUnit tests across successive stages from refund to financial notification, verifying business error handling, system retries, status-update failures, and recovery.

DataAnnotation | AI Trainer — Software Engineering Oct 2026 – Present
- Design challenging software engineering tasks to expose coding agent failures; develop reference implementations and automated test suites for implementation-based evaluations.
- Evaluate agent reasoning and generated solutions against task requirements, identifying subtle logic errors, flawed implementation decisions, missed edge cases, and unreliable tests.

Projects

Urbanova | ***Python, TypeScript, React, Mapbox, Express, PostgreSQL*** 2026
Hack the Valley: See how proposed Toronto housing could change neighbourhood views, sunlight, and streetcar commutes.
- Led integration from the Python simulation through Express to React, defining data contracts and coordinating schema changes to keep 3D views synchronized with simulation results.
- Built the 505 Dundas transit simulation, modelling passenger trips from residential areas to workplace destinations, streetcar frequency and headways, variable delays, and crowding.
- Developed a 3D replay, driven by recorded simulation events, showing moving streetcars, riders walking to stops, passenger counts, and crowding on the interactive map with data visualizations.

Tempo | ***React, TypeScript, Node.js, Express, MongoDB, Better Auth, Railway*** 2026–Present
Full-stack calendar focused on making scheduling faster and more intuitive through simpler interactions.
- Independently designed, developed, and deployed Tempo end to end, implementing account authentication and user-scoped event storage so users can access their own calendars across sessions and devices.
- Engineered a regex parser that outperforms Google and Apple Calendar across 3+ tested scheduling scenarios, recognizing multi-day and combined date-time ranges (with "now" endpoints) in one field.
- Devised a one-click time entry workflow with instant AM/PM toggling for faster event creation.

GirlMath | ***Python, FastAPI, React Native, TypeScript, MongoDB, DigitalOcean, Gemini API*** 2026
AI powered journaling app supporting girls in STEM by tracking confidence, motivation, and learning trends.
- Led backend dev in a 36-hour hackathon, built FastAPI logic and REST endpoints for a 4 person team.
- Integrated the Gemini API to derive confidence, motivation, and learning trends from journal entries.
- Built frontend API client for backend communication and deployed the FastAPI service to DigitalOcean.

Skills

- **Programming Languages:** Java, Python, TypeScript/JavaScript, SQL, C, HTML/CSS, Assembly
- **Development Stack:** Spring Boot, React/React Native, Node.js, Express.js, Agents/Codex, FastAPI, Expo, Java Swing, JUnit, Maven
- **Data & Tools:** PostgreSQL, MongoDB, AWS, Azure, Git, Linux, DigitalOcean, Railway
- **Languages:** Proficient in English, French, and Chinese